To access an Oracle 19c database from IBM Db2 10.5.0.3 hosted on an AIX server, you must configure Db2 Federation. The most robust and feature-rich method is using the native Oracle client wrapper (NET8).

Below is the step-by-step process to complete this setup.

Step 1: Install and Configure the Oracle Client on AIX with user oracle and permissions:

Add db2isur in oracle group permission like dba,oinstall,db2iusr.

The Db2 instance owner must be able to resolve and connect to the remote Oracle 19c database using Oracle client libraries.

1. Install Oracle Client: Install a 64-bit Oracle 19c Client (or Oracle Instant Client) on the AIX server.

2. Configure tnsnames.ora: Create or edit the $ORACLE_HOME/network/admin/tnsnames.ora file to include the connection string for your Oracle 19c database:text

```
ORACLE19C =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = oracle_host_ip)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = oracle_service_name)
    )
  )
```

Use code with caution.

3. Verify Connectivity: Log in as the Db2 instance owner (db2iusr) and verify that you can ping/connect to Oracle:bash

```
sqlplus oracle_user/oracle_password@ORACLE19C
```

Use code with caution.

OUT-PUT:

```
$ sqlplus sys/sys123@AYUSH as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Mon Oct 5 12:23:53 2026
Version 19.3.0.0.0

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Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL>
```

Step 2: Configure Db2 Environment Variables (db2dj.ini)

Db2 needs to know where the Oracle home directory and network configuration are located.

1. Log in as the Db2 instance owner on AIX.

2. Edit OR create the db2dj.ini file located in your instance directory (typically ~db2inst1/sqllib/cfg/db2dj.ini). Add the following environment paths:ini

ORACLE_HOME=/u01/app/oracle/product/19.0.0/client_1

TNS_ADMIN=/u01/app/oracle/product/19.0.0/client_1/network/admin

Use code with caution.

OUT-PUT:

```
$ cat db2dj.ini
ORACLE_HOME=/db2/oracle/product/19.0.0/client_1/client/product/19.0.0/client_1
TNS_ADMIN=/db2/oracle/product/19.0.0/client_1/client/product/19.0.0/client_1/network/admin
```

3. (Optional) Update your user .profile or db2profile to match these variables.

Step 3: Enable Federation and Restart Db2

Federation must be turned on at the Db2 instance level.

1. Turn on the federated parameter:bash

db2 update dbm cfg using FEDERATED YES

Use code with caution.

OUT-PUT:

```
$ db2 get dbm cfg | grep -i fed
 Federated Database System Support           (FEDERATED) = YES
 Bypass federated authentication            (FED_NOAUTH) = NO
 Maximum Asynchronous TQs per query    (FEDERATED_ASYNC) = 0
$
```

2. Restart the Db2 instance to apply both the db2dj.ini paths and the configuration change:bash

db2stop

db2start

Use code with caution.

Step 4: Create Federated Objects in Db2

Connect to your local Db2 database via the command line interface to register the Oracle wrapper, server definition, and user mappings.

1. Register the Wrapper

Register the native Oracle client wrapper. When using the default name NET8, Db2 automatically points to the underlying AIX library (libdb2net8.a):

sql

db2 "CREATE WRAPPER NET8"

Use code with caution.

OUT-PUT:

```
$ db2 "CREATE WRAPPER NET8"
DB20000I  The SQL command completed successfully.
```

SELECT WRAPNAME, WRAPTYPE, LIBRARY FROM SYSCAT.WRAPPERS;

```
$ db2 "SELECT WRAPNAME, WRAPTYPE, LIBRARY FROM SYSCAT.WRAPPERS"

WRAPNAME                                                
```

Define the remote Oracle 19c server. The NODE parameter must match the exact alias defined in your tnsnames.ora file.

sql

db2 "CREATE SERVER AYUSH

TYPE ORACLE

VERSION 19

WRAPPER NET8

OPTIONS (NODE 'AYUSH')"

Use code with caution.

OUT-PUT:

```
$ db2 "CREATE SERVER AYUSH
      TYPE ORACLE
      VERSION 19.3
      WRAPPER NET8
      OPTIONS (NODE 'AYUSH')"
> > > > SQL1417W  Wrapper "NET8" supports data source server versions "8.x, 9.x, 10.x,
11.x, 12.x".  Using the wrapper with later versions might result in errors or
unexpected results.  SQLSTATE=0168Q
```

SELECT SERVERNAME, WRAPNAME, SERVERTYPE, SERVERVERSION FROM SYSCAT.SERVERS;

```
$ db2 "SELECT SERVERNAME, WRAPNAME, SERVERTYPE, SERVERVERSION
    FROM SYSCAT.SERVERS">

SERVERNAME                                                                                                                       WRAPNAME
                                                SERVERTYPE                  SERVERVERSION
-------------------------------------------------------------------------------------------------------------------------------- --------------
------------------------------------------------ ---------------------------- -----------------
TGTSOLIXDB                                                                                                                       DRDA
                                                DB2/UDB                     10.5
AYUSH                                                                                                                            NET8
                                                ORACLE                      19.3
```

NOTE: WE HAVE TO CREATE A DATABASE USER(db2_test1) IN ORACLE AND GRANT NECESSARY PERMISSIONS TO CONNECT DATABASE AND SELECT TABLES.

3. Create the User Mapping

Map your local Db2 authorization user to the remote Oracle credential authorized to select the target tables.

sql

db2 "CREATE USER MAPPING FOR db2iusr

SERVER AYUSH

OPTIONS (REMOTE_AUTHID 'oracle_db_user', REMOTE_PASSWORD 'oracle_password')"

Use code with caution.

OUT-PUT:

```
$ db2 "CREATE USER MAPPING FOR db2iusr
     SERVER AYUSH
     OPTIONS (SET REMOTE_AUTHID 'db2_test1', SET REMOTE_PASSWORD 'db2_test1#123')"> >
DB20000I  The SQL command completed successfully.
```

SELECT SUBSTR(AUTHID, 1, 15) AS AUTHID,AUTHIDTYPE,SUBSTR(SERVERNAME, 1, 15) AS SERVERNAME,SUBSTR(OPTION, 1, 20) AS OPTION,SUBSTR(SETTING, 1, 20) AS SETTING FROM SYSCAT.USEROPTIONS WHERE SERVERNAME = 'AYUSH';

```
$ db2 "SELECT SUBSTR(AUTHID, 1, 15) AS AUTHID, AUTHIDTYPE, SUBSTR(SERVERNAME, 1, 15) AS SERVERNAME, SU
AME = 'AYUSH'"

AUTHID          AUTHIDTYPE SERVERNAME      OPTION              SETTING
--------------- ---------- --------------- ------------------- --------------------
DB2IUSR         U          AYUSH           REMOTE_AUTHID       db2_test1
DB2IUSR         U          AYUSH           REMOTE_PASSWORD     ********
```

Passthrough Testing: Using `SET PASSTHRU AYUSH` and querying `DUAL` to verify network and credentials.

```
$ db2 "SET PASSTHRU AYUSH"
DB20000I  The SQL command completed successfully.
$ db2 "SELECT * FROM dual"

DUMMY
-----
X

  1 record(s) selected.

$ db2 "SET PASSTHRU RESET"
DB20000I  The SQL command completed successfully.
```

Step 5: Map and Access Oracle Tables (Nicknames)

To interact with an Oracle table from Db2, you create a local pointer called a Nickname.

1. Create a nickname for your specific Oracle table:sql

db2 "CREATE NICKNAME table_nickname

FOR ORA19CSERVER.ORACLE_SCHEMA.REMOTE_TABLE_NAME"

Use code with caution.

OUT-PUT:

```
$ db2 "CREATE NICKNAME TEST_ORACLE_1 FOR AYUSH.VIJAY.RESULT1"
DB20000I  The SQL command completed successfully.
```

SELECT TABSCHEMA, TABNAME, SERVERNAME, REMOTE_SCHEMA, REMOTE_TABLE FROM SYSCAT.NICKNAMES;

```
$ db2 "SELECT TABSCHEMA, TABNAME, SERVERNAME, REMOTE_SCHEMA, REMOTE_TABLE
     FROM SYSCAT.NICKNAMES">

TABSCHEMA
                                                        SERVERNAME


------------------------------------------------------------------------------------------------------------
------------------------------------------------- ------------------------------------------------------
------------------------------------------------------------------------------------------------------------
--------------------------------
DB2IUSR
                                                        AYUSH


DB2IUSR
                                                        AYUSH
```

2. Query the data from Db2 exactly as if it were a local table:sql

db2 "SELECT * FROM db2schema.local_table_alias FETCH FIRST 10 ROWS ONLY"

Use code with caution.

OUT-PUT: ON DB2

```
$ db2 connect to sblprddb

   Database Connection Information

 Database server        = DB2/AIX64 10.5.10
 SQL authorization ID   = DB2IUSR
 Local database alias   = SBLPRDDB

$ db2 "SELECT * FROM TEST_ORACLE_1 FETCH FIRST 10 ROWS ONLY"

CASE_NUMBER                                NHPM_ID             FAMILY_ID
      SPECIALITY_CODE
------------------------------------------ ------------------- ------------------------------------------
----- ---------------
CASE/PS6/HOSP1P98893/JK1091377             PVNIUTUK1           1JK0106007001500000003600043
      MO
CASE/PS6/HOSP1P48968/JK1065082             PXGELX8YV           1JK0116004010500000014600041
      SL
CASE/PS6/HOSP1P144083/JK1118387            P30MKOEJ8           1JK0120002015200000021100085
      SE
CASE/PS6/HOSP1P134391/JK1080205            PJIJUVXK8           1JK0110001703300501889000013
      SG
CASE/PS6/HOSP1P114724/JK1091733            P48T5R545           1JK0122001708500020003000036
      MG
CASE/PS6/HOSP1P00511/S1129714              PGVESQ2WM           1S0106005002400000003500094
      MC
CASE/PS6/HOSP1P114724/JK1084556            P4526EISX           1JK0119001706700110044000020
      MG
CASE/PS6/HOSP1P108365/NF1083346            P8VFU6TQM           1NF011000450647
      MG
CASE/PS6/HOSP1P124017/JK1080838            P0RF0SLFY           1JK0121002707700931258000024
      MG
CASE/PS6/HOSP1P132114/NF1079836            PNA46YUVT           1NF011001155723
      MG

  10 record(s) selected.
```

OUT-PUT: ON ORACLE

```
SELECT * FROM VIJAY.RESULT1 FETCH FIRST 10 ROWS ONLY;
```

Script Output x | Query... x

SQL | All Rows Fetched: 10 in 0.003 seconds

	CASE_NUMBER	NHPM_ID	FAMILY_ID	PATIENT_NAME	PRO
1	CASE/PS6/HOSP1P98893/JK1091377	PVNIUTUK1	1JK0106007001500000003600043	Anku Sharma	MO024
2	CASE/PS6/HOSP1P48968/JK1065082	PXGELX8YV	1JK0116004010500000014600041	Farhana Bano	SL004
3	CASE/PS6/HOSP1P144083/JK1118387	P30MKOEJ8	1JK0120002015200000021100085	Rekhi Ram	SE020
4	CASE/PS6/HOSP1P134391/JK1080205	PJIJUVXK8	1JK0110001703300501889000013	Mohammad Musaib Kakroo	SG039
5	CASE/PS6/HOSP1P114724/JK1091733	P48T5R545	1JK0122001708500020003000036	PUSHPA SHARMA	MG038
6	CASE/PS6/HOSP1P00511/S1129714	PGVESQ2WM	1S0106005002400000003500094	Lal Hussain	MC011
7	CASE/PS6/HOSP1P114724/JK1084556	P4526EISX	1JK0119001706700110044000020	OM PARKASH	MG072
8	CASE/PS6/HOSP1P108365/NF1083346	P8VFU6TQM	1NF011000450647	Irshad Ahmad Kundoo	MG072
9	CASE/PS6/HOSP1P124017/JK1080838	P0RF0SLFY	1JK0121002707700931258000024	Bansi Singh	MG072
10	CASE/PS6/HOSP1P132114/NF1079836	PNA46YUVT	1NF011001155723	Abdul Hamid Butt	MG072